SETK 4643

Thermodynamics

Chapter 9: Gas Power Cycles

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9.1 Basic Considerations in the Analysis of Power Cycles

- Our study of gas power cycles will involve the study of those heat engines in which the working fluid remains in the gaseous state throughout the cycle.
- When the actual cycle is stripped of all the internal irreversibilities and complexities, it is called an ideal cycle.
- The ideal cycle resembles the actual cycle closely but is made up totally of internally reversible processes.
- Reversible cycles such as Carnot cycle have the highest thermal efficiency of all heat engines operating between the same temperature levels.
- Unlike ideal cycles, they are totally reversible, and unsuitable as a realistic model.

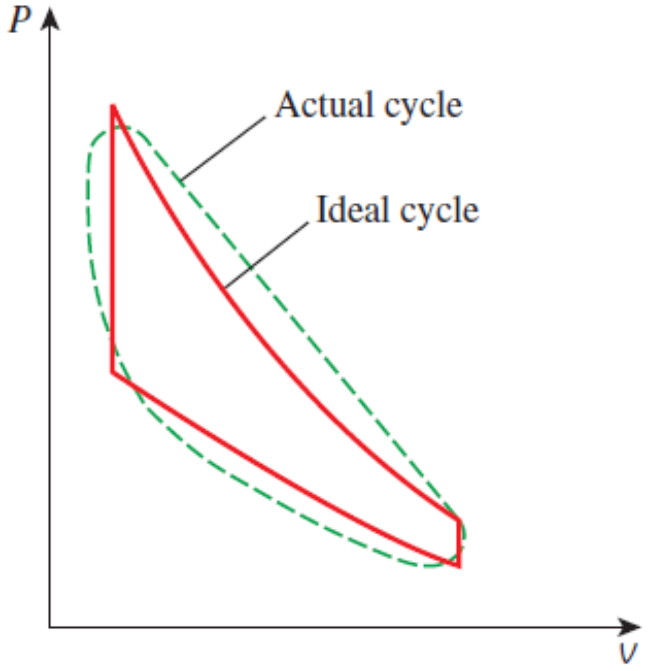
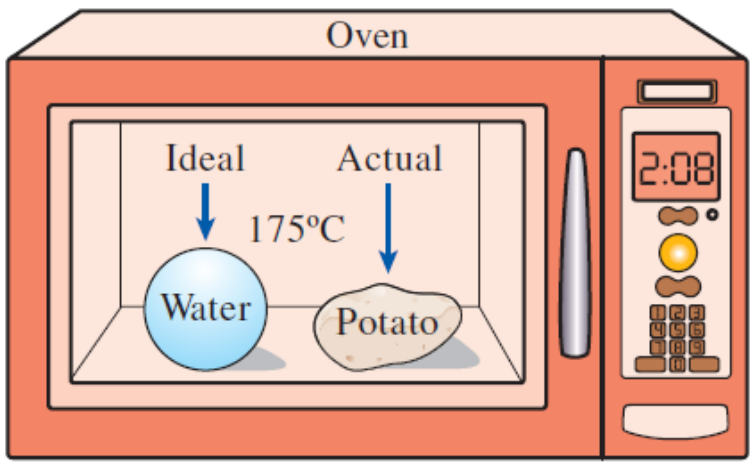


FIGURE 9–2
The analysis of many complex processes can be reduced to a manageable level by utilizing some idealizations.

9.1 Basic Considerations in the Analysis of Power Cycles (2)

- The ideal cycles are internally reversible, but, unlike the Carnot cycle, they are not necessarily externally reversible.
- Therefore, the thermal efficiency of an ideal cycle, in general, is less than that of a totally reversible cycle operating between the same temperature limits.
- However, it is still considerably higher than the thermal efficiency of an actual cycle because of the idealizations utilized.
- We will be concerned with how the major parameters of the cycle affect the performance of heat engines.
- The performance is often measured in terms of the cycle efficiency.

$$\eta_{th} = \frac{W_{net}}{Q_{in}} \qquad (9.1)$$



Modeling is a powerful engineering tool that provides great insight and simplicity at the expense of some loss in accuracy.

9.1 Basic Considerations in the Analysis of Power Cycles (2)

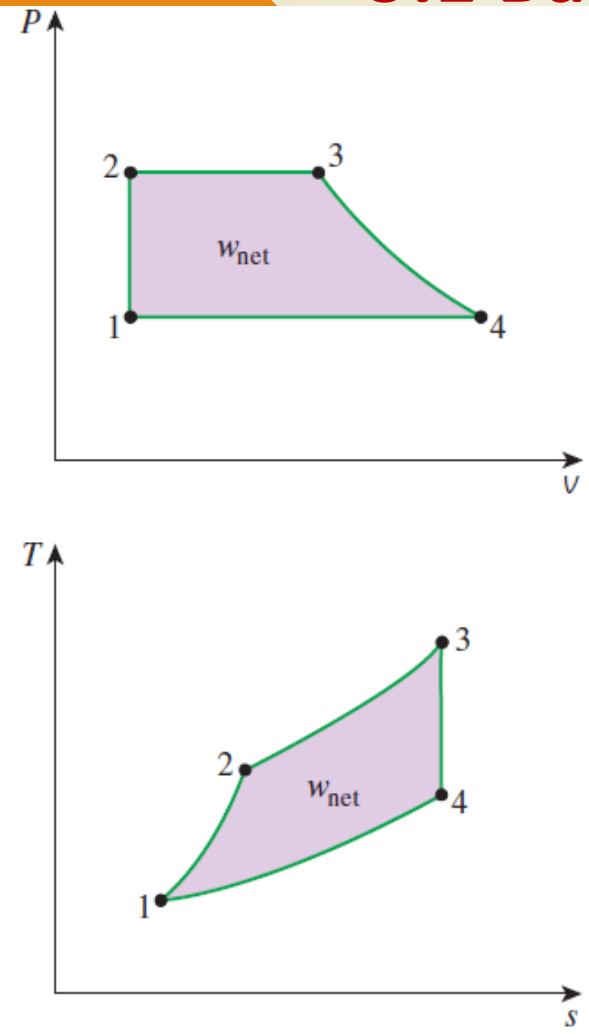


FIGURE 9–4
On both *P*-*v* and *T*-*s* diagrams, the area enclosed by the process curve represents the net work of the cycle.

- The idealizations and simplifications in the analysis of power cycles:
  1. The cycle does not involve any friction. Therefore, the working fluid does not experience any pressure drop as it flows in pipes or devices such as heat exchangers.
  2. All expansion and compression processes take place in a quasi-equilibrium manner.
  3. The pipes connecting the various components of a system are well insulated, and heat transfer through them is negligible.

- On a *T*-*s* diagram, the ratio of the area enclosed by the cyclic curve to the area under the heat-addition process curve represents the thermal efficiency of the cycle.
- Any modification that increases the ratio of these two areas will also increase the thermal efficiency of the cycle.

9.2 The Carnot Cycle

- Heat engines that operate on a totally reversible cycle, such as the Carnot cycle, have the highest thermal efficiency of all heat engines operating between the same temperature levels.
- The Carnot cycle is described by the following four processes:

Process	Description
1-2	Isothermal Heat Addition
2-3	Isentropic Expansion
3-4	Isothermal Heat Rejection
4-1	Isentropic Compression

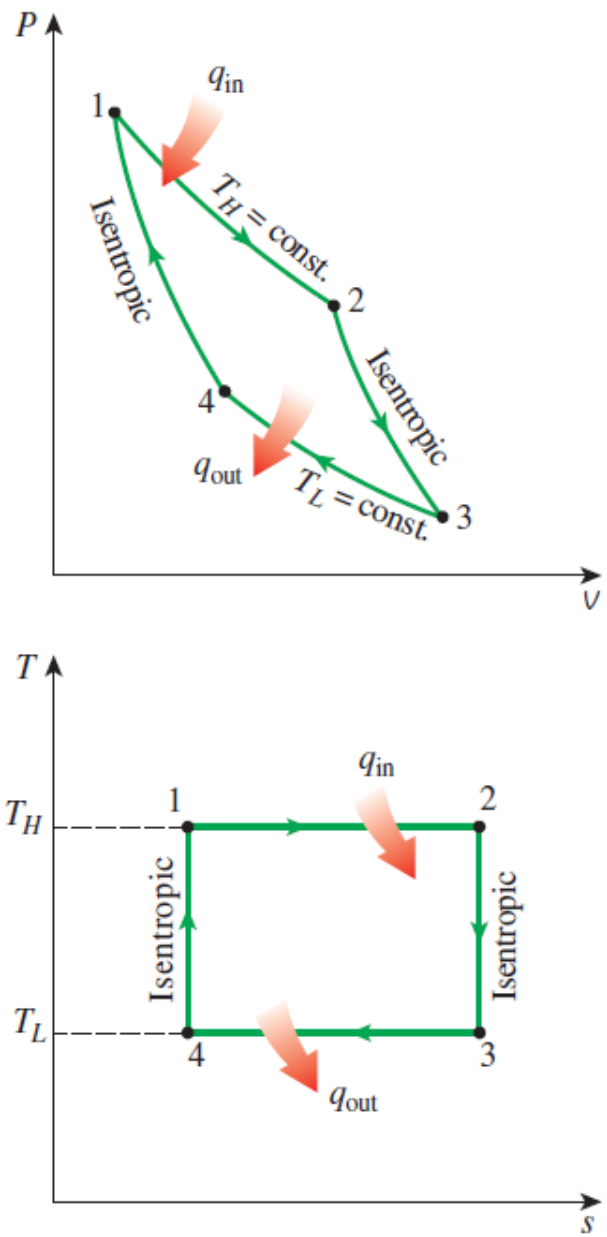


FIGURE 9–5
P-v and *T-s* diagrams of a Carnot cycle.

9.2 The Carnot Cycle (2)

- The areas under the process curves on the *P-v* diagram represent the work done for closed systems. The net cycle work done is the area enclosed by the cycle on the *P-v* diagram.
- The areas under the process curves on the *T-s* diagram represent the heat transfer for the processes. The net heat added to the cycle is the area that is enclosed by the cycle on the *T-s* diagram.
- For a cycle, $W_{net} = Q_{net}$; therefore, the areas enclosed on the *P-v* and *T-s* diagrams are equal.

$$\eta_{th,\,Carnot} = 1 - \frac{T_L}{T_H} \qquad (9.2)$$

- Thermal efficiency increases with an increase in the average temperature at which heat is supplied to the system or with a decrease in the average temperature at which heat is rejected from the system.

Example 1

EXAMPLE 9–1 **Derivation of the Efficiency of the Carnot Cycle**

Show that the thermal efficiency of a Carnot cycle operating between the temperature limits of T_H and T_L is solely a function of these two temperatures and is given by Eq. 9–2.

SOLUTION It is to be shown that the efficiency of a Carnot cycle depends on the source and sink temperatures alone.

Analysis The T-s diagram of a Carnot cycle is redrawn in Fig. 9–7. All four processes that comprise the Carnot cycle are reversible, and thus the area under each process curve represents the heat transfer for that process. Heat is transferred to the system during process 1-2 and rejected during process 3-4. Therefore, the amount of heat input and heat output for the cycle can be expressed as

$$q_{in} = T_H(s_2 - s_1) \quad \text{and} \quad q_{out} = T_L(s_3 - s_4) = T_L(s_2 - s_1)$$

since processes 2-3 and 4-1 are isentropic, and thus $s_2 = s_3$ and $s_4 = s_1$. Substituting these into Eq. 9–1, we see that the thermal efficiency of a Carnot cycle is

$$\eta_{th} = \frac{w_{net}}{q_{in}} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{T_L(s_2 - s_1)}{T_H(s_2 - s_1)} = 1 - \frac{T_L}{T_H}$$

Discussion Notice that the thermal efficiency of a Carnot cycle is independent of the type of the working fluid used (an ideal gas, steam, etc.) or whether the cycle is executed in a closed or steady-flow system.

9.3 Air-Standard Assumptions

- Air-standard assumptions reduce the analysis of gas power cycles to a manageable level by utilizing the following approximations:
  - The working fluid is air, which continuously circulates in a closed loop and always behaves as an ideal gas.
  - All the processes that make up the cycle are internally reversible.
  - The combustion process is replaced by a heat-addition process from an external source.
  - The exhaust process is replaced by a heat rejection process that restores the working fluid to its initial state.
- The cold-air-standard assumptions apply when the working fluid is air and has constant specific heat evaluated at room temperature (25°C).
- **Air-standard cycle**: A cycle for which the air-standard assumptions are applicable.

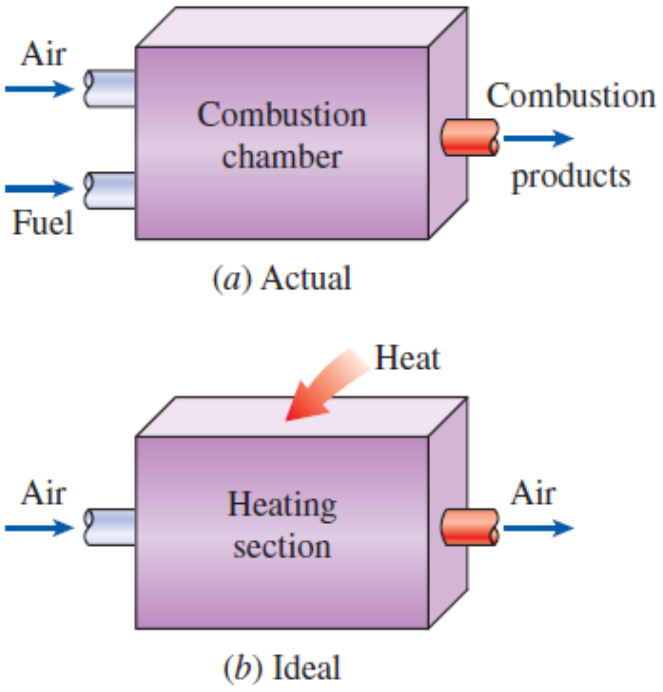


FIGURE 9–8
The combustion process is replaced by a heat-addition process in ideal cycles.

Example 2

EXAMPLE 9–2 An Air-Standard Cycle

An air-standard cycle is executed in a closed system and is composed of the following four processes:

1-2 Isentropic compression from 100 kPa and 27°C to 1 MPa
2-3 $P = constant$ heat addition in amount of 2800 kJ/kg
3-4 $v = constant$ heat rejection to 100 kPa
4-1 $P = constant$ heat rejection to initial state

(*a*) Show the cycle on *P-v* and *T-s* diagrams.
(*b*) Calculate the maximum temperature in the cycle.
(*c*) Determine the thermal efficiency.

Assume constant specific heats at room temperature.

SOLUTION The four processes of an air-standard cycle are described. The cycle is to be shown on *P-v* and *T-s* diagrams, and the maximum temperature in the cycle and the thermal efficiency are to be determined.

Assumptions **1** The air-standard assumptions are applicable. **2** Kinetic and potential energy changes are negligible. **3** Air is an ideal gas with constant specific heats.

Properties The properties of air at room temperature are $c_p = 1.005$ kJ/kg·K, $c_v = 0.718$ kJ/kg·K, and $k = 1.4$ (Table A–2*a*).

Analysis (*a*) The cycle is shown on *P-v* and *T-s* diagrams in Fig. 9–9.

(*b*) From the ideal gas isentropic relations and energy balance,

$$T_2 = T_1\left(\frac{P_2}{P_1}\right)^{(k-1)/k} = (300\text{ K})\left(\frac{1000\text{ kPa}}{100\text{ kPa}}\right)^{0.4/1.4} = 579.2\text{ K}$$

$$q_{in} = h_3 - h_2 = c_p(T_3 - T_1)$$

$$2800\text{ kJ/kg} = (1.005\text{ kJ/kg·K})(T_3 - 579.2) \longrightarrow T_{max} = T_3 = \mathbf{3360\text{ K}}$$

(*c*) The temperature at state 4 is determined from the ideal gas relation for a fixed mass,

$$\frac{P_3 v_3}{T_3} = \frac{P_4 v_4}{T_4} \longrightarrow T_4 = \frac{P_4}{P_3}T_3 = \frac{100\text{ kPa}}{1000\text{ kPa}}(3360\text{ K}) = 336\text{ K}$$

The total amount of heat rejected from the cycle is

$$\begin{aligned} q_{out} &= q_{34,out} + q_{41,out} = (u_3 - u_4) + (h_4 + h_1) \\ &= c_v(T_3 - T_4) + c_p(T_4 - T_1) \\ &= (0.718\text{ kJ/kg·K})(3360 - 336)\text{K} + (1.005\text{ kJ/kg·K})(336 - 300)\text{K} \\ &= 2212\text{ kJ/kg} \end{aligned}$$

Then, the thermal efficiency is determined from its definition to be

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{2212\text{ kJ/kg}}{2800\text{ kJ/kg}} = 0.210 \text{ or } \mathbf{21.0\%}$$

Discussion The assumption of constant specific heats at room temperature is not realistic in this case since the temperature changes involved are too large.

9.4 An Overview of Reciprocating Engines

- The basic components of a reciprocating engine are shown in Fig. 9.4.
- The piston reciprocates in the cylinder between two fixed positions called the top dead center (TDC) and the bottom dead center (BDC).
- **Stroke** is the distance between the top dead center and the bottom dead center and is the largest distance that the piston can travel in one direction within a cylinder.
- **Bore** is the diameter of the piston.
- The ratio of the maximum volume formed in the cylinder to the minimum (clearance) volume is called the **compression ratio *r*** of the engine:

$$r = \frac{V_{\max}}{V_{\min}} = \frac{V_{BDC}}{V_{TDC}} \qquad (9.3)$$

Reading assignment

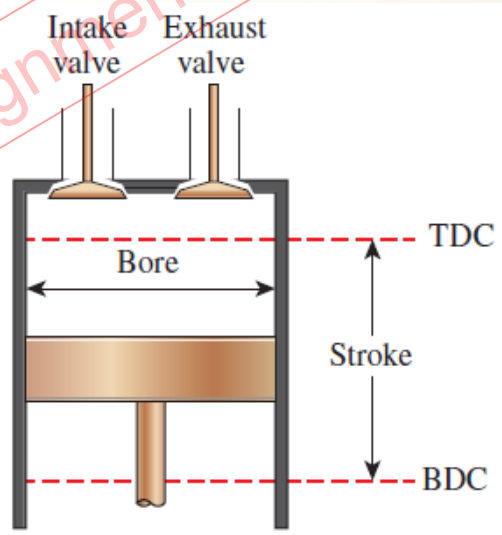


FIGURE 9–9
Nomenclature for reciprocating engines.

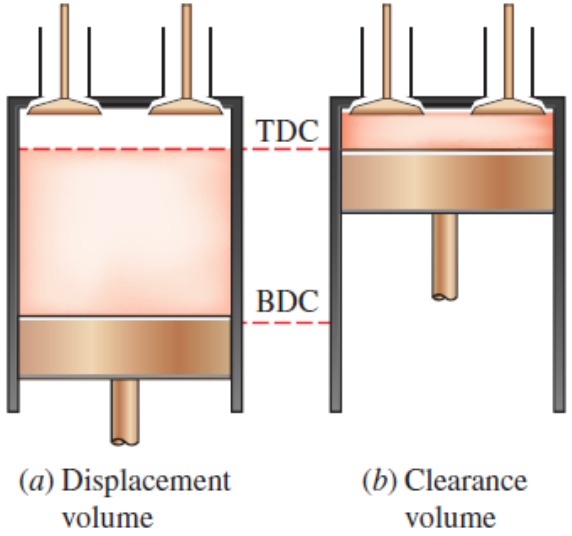


FIGURE 9–10
Displacement and clearance volumes of a reciprocating engine.

9.4 An Overview of Reciprocating Engines (2)

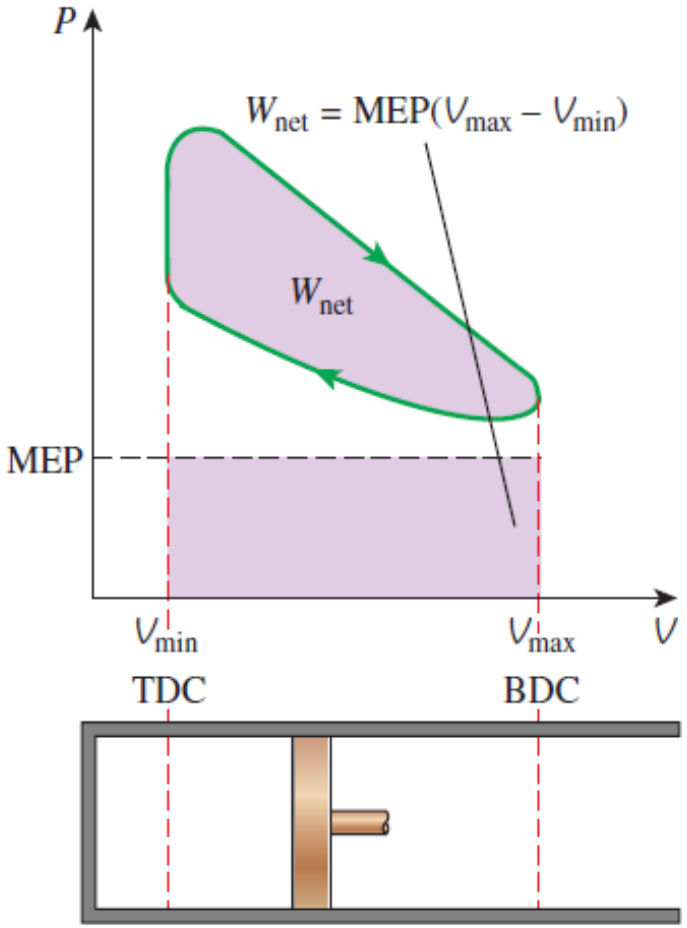


FIGURE 9–11
The net work output of a cycle is equivalent to the product of the mean effective pressure and the displacement volume.

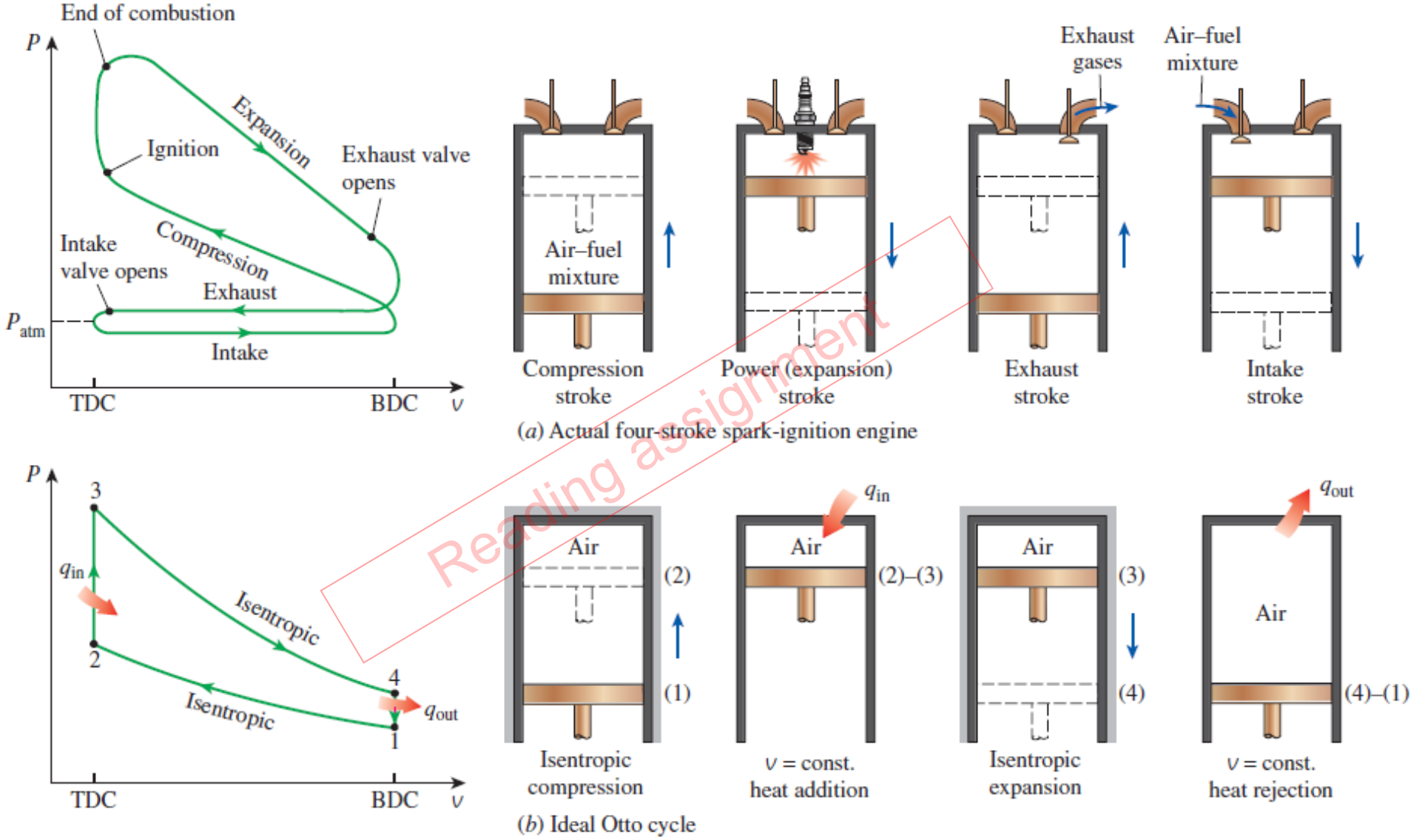
- **Mean effective pressure (MEP)** is a fictitious pressure that, if it acted on the piston during the entire power stroke, would produce the same amount of net work as that produced during the actual cycle.
- $\boldsymbol{W_{net}}$ **=** ***MEP*** **x Piston area x Stroke =** ***MEP*** **x Displacement volume,**

or $$MEP = \frac{W_{net}}{V_{max} - V_{min}} = \frac{w_{net}}{v_{max} - v_{min}} \quad \text{(kPa)}$$

Reading assignment

- The mean effective pressure can be used as a parameter to compare the performances of reciprocating engines of equal size.
- The engine with a larger value of MEP delivers more net work per cycle and thus performs better.
- Reciprocating engines are classified as **spark-ignition (SI)** engines or **compression-ignition (CI)** engines, depending on how the combustion process in the cylinder is initiated.

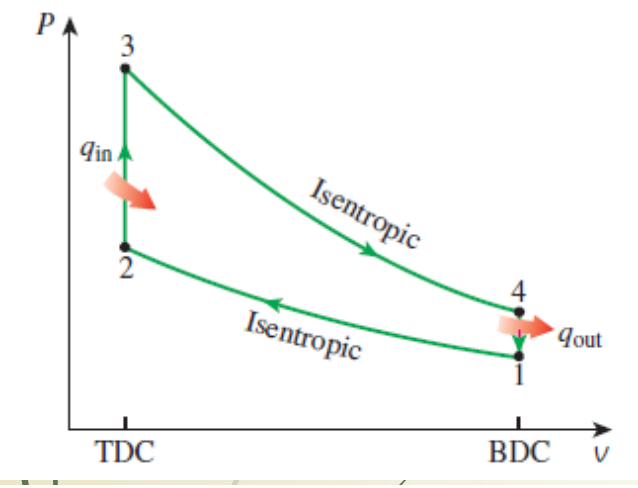
9.5 Otto Cycle: The Ideal Cycle for Spark-ignition Engines



Reading assignment

FIGURE 9–12
Actual and ideal cycles in spark-ignition engines and their P-v diagrams.

9.5 Otto Cycle: The Ideal Cycle for Spark-ignition Engines (2)



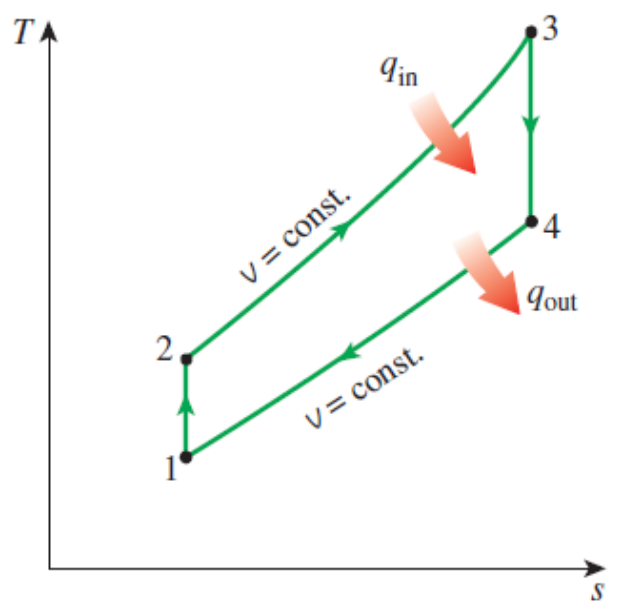


FIGURE 9–15

- The Air-Standard Otto Cycle is the ideal cycle that approximates the spark-ignition combustion engine

Process	Description
1-2	Isentropic Compression
2-3	Constant-Volume Heat Addition
3-4	Isentropic Expansion
4-1	Constant-Volume Heat Rejection

- No work is involved during the two heat transfer processes since both take place at const. volume.
- Therefore, heat transfer to and from the working fluid can be expressed as

$$q_{in} = u_3 - u_2 = c_v(T_3 - T_2) \qquad (9.6a)$$

$$q_{out} = u_4 - u_1 = c_v(T_4 - T_1) \qquad (9.6a)$$

9.5 Otto Cycle: The Ideal Cycle for Spark-ignition Engines (3)

- The thermal efficiency of the ideal Otto cycle under the cold air standard assumptions can be expressed as

$$\eta_{th,Otto} = \frac{w_{net}}{q_{in}} = 1 - \frac{q_{out}}{q_{in}}$$

$$= 1 - \frac{T_4 - T_1}{T_3 - T_2} = 1 - \frac{T_1(T_4/T_1 - 1)}{T_2(T_3/T_2 - 1)}$$

- Processes 1-2 and 3-4 are isentropic, and $v_2 = v_3$ and $v_4 = v_1$. Thus,

$$\frac{T_1}{T_2} = \left(\frac{v_2}{v_1}\right)^{k-1} = \left(\frac{v_3}{v_4}\right)^{k-1} = \frac{T_4}{T_3} \qquad (9.7)$$

$$\therefore \ \eta_{th,Otto} = 1 - \frac{1}{r^{k-1}} \qquad (9.8) \quad \text{where} \qquad r = \frac{V_{max}}{V_{min}} = \frac{V_1}{V_2} = \frac{v_1}{v_2} \qquad (9.9)$$

r is the compression ratio and *k* is the specific heat ratio c_p/c_v.

9.5 Otto Cycle: The Ideal Cycle for Spark-ignition Engines (4)

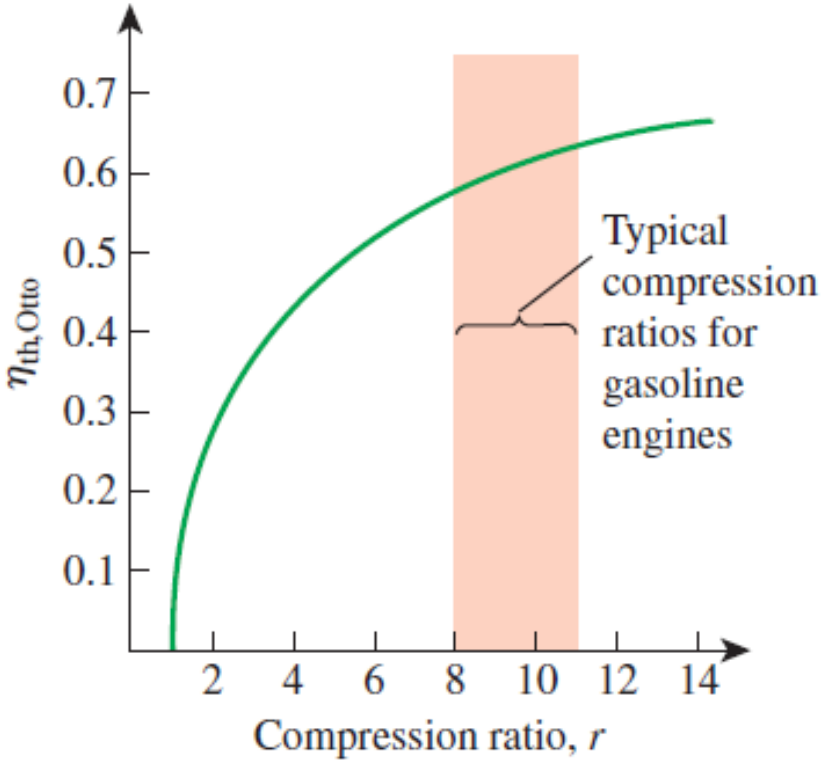


FIGURE 9–17
Thermal efficiency of the ideal Otto cycle as a function of compression ratio (k = 1.4).

- k = 1.4 – specific heat ratio of air at room temperature
- k = 1.667 – specific heat ratio of argon or helium
- Carbon dioxide – k = 1.3
- Ethane – k = 1.2
- k decreases with increase in temperature
- Thermal efficiency of actual spark-ignition engines range from 25 -30 percent

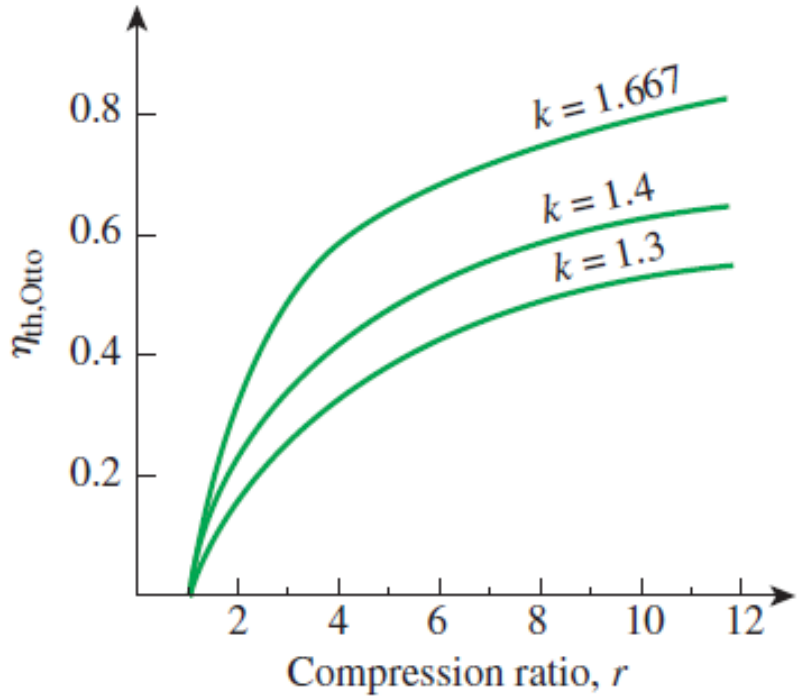


FIGURE 9–18
The thermal efficiency of the Otto cycle increases with the specific heat ratio k of the working fluid.

Example 3

EXAMPLE 9–3 The Ideal Otto Cycle

An ideal Otto cycle has a compression ratio of 8. At the beginning of the compression process, air is at 100 kPa and 17°C, and 800 kJ/kg of heat is transferred to air during the constant-volume heat-addition process. Accounting for the variation of specific heats of air with temperature, determine (*a*) the maximum temperature and pressure that occur during the cycle, (*b*) the net work output, (*c*) the thermal efficiency, and (*d*) the mean effective pressure for the cycle.

(*e*) Also, determine the power output from the cycle, in kW, for an engine speed of 4000 rpm (rev/min). Assume that this cycle is operated on an engine that has four cylinders with a total displacement volume of 1.6 L.

SOLUTION An ideal Otto cycle is considered. The maximum temperature and pressure, the net work output, the thermal efficiency, the mean effective pressure, and the power output for a given engine speed are to be determined.

Assumptions **1** The air-standard assumptions are applicable. **2** Kinetic and potential energy changes are negligible. **3** The variation of specific heats with temperature is to be accounted for.

Analysis The *P*-*v* diagram of the ideal Otto cycle described is shown in Fig. 9–20. We note that the air contained in the cylinder forms a closed system.

(*a*) The maximum temperature and pressure in an Otto cycle occur at the end of the constant-volume heat-addition process (state 3). But first we need to determine the temperature and pressure of air at the end of the isentropic compression process (state 2), using data from Table A–17:

$$T_1 = 290\text{ K} \rightarrow u_1 = 206.91\text{ kJ/kg}$$
$$v_{r1} = 676.1$$

Process 1-2 (isentropic compression of an ideal gas):

$$\frac{v_{r2}}{v_{r1}} = \frac{v_2}{v_1} = \frac{1}{r} \rightarrow v_{r2} = \frac{v_{r1}}{r} = \frac{676.1}{8} = 84.51 \rightarrow T_2 = 652.4\text{ K}$$
$$u_2 = 475.11\text{ kJ/kg}$$

$$\frac{P_2 v_2}{T_2} = \frac{P_1 v_1}{T_1} \rightarrow P_2 = P_1\left(\frac{T_2}{T_1}\right)\left(\frac{v_1}{v_2}\right)$$
$$= (100\text{ kPa})\left(\frac{652.4\text{ K}}{290\text{ K}}\right)(8) = 1799.7\text{ kPa}$$

Process 2-3 (constant-volume heat addition):

$$q_{in} = u_3 - u_2$$
$$800\text{ kJ/kg} = u_3 - 475.11\text{ kJ/kg}$$
$$u_3 = 1275.11\text{ kJ/kg} \rightarrow T_3 = \mathbf{1575.1\text{ K}}$$
$$v_{r3} = 6.108$$

$$\frac{P_3 v_3}{T_3} = \frac{P_2 v_2}{T_2} \rightarrow P_3 = P_2\left(\frac{T_3}{T_2}\right)\left(\frac{v_2}{v_3}\right)$$
$$= (1.7997\text{ MPa})\left(\frac{1575.1\text{ K}}{652.4\text{ K}}\right)(1) = \mathbf{4.345\text{ MPa}}$$

Example 3

(*b*) The net work output for the cycle is determined either by finding the boundary ($P\,dV$) work involved in each process by integration and adding them or by finding the net heat transfer that is equivalent to the net work done during the cycle. We take the latter approach. However, first we need to find the internal energy of the air at state 4:

Process 3-4 (isentropic expansion of an ideal gas):

$$\frac{v_{r4}}{v_{r3}} = \frac{v_4}{v_3} = r \rightarrow v_{r4} = rv_{r3} = (8)(6.108) = 48.864 \quad \rightarrow \quad T_4 = 795.6 \text{ K}$$

$$u_4 = 588.74 \text{ kJ/kg}$$

Process 4-1 (constant-volume heat rejection):

$$-q_{out} = u_1 - u_4 \rightarrow q_{out} = u_4 - u_1$$

$$q_{out} = 588.74 - 206.91 = 381.83 \text{ kJ/kg}$$

Thus,

$$w_{net} = q_{net} = q_{in} - q_{out} = 800 - 381.83 = \mathbf{418.17\ kJ/kg}$$

(*c*) The thermal efficiency of the cycle is determined from its definition:

$$\eta_{th} = \frac{w_{net}}{q_{in}} = \frac{418.17 \text{ kJ/kg}}{800 \text{ kJ/kg}} = 0.523 \quad \text{or} \quad \mathbf{52.3\%}$$

Under the cold-air-standard assumptions (constant specific heat values at room temperature), the thermal efficiency would be (Eq. 9–8)

$$\eta_{th,Otto} = 1 - \frac{1}{r^{k-1}} = 1 - r^{1-k} = 1 - (8)^{1-1.4} = 0.565 \quad \text{or} \quad 56.5\%$$

which is considerably different from the value obtained above. Therefore, care should be exercised in utilizing the cold-air-standard assumptions.

(*d*) The mean effective pressure is determined from its definition, Eq. 9–4:

$$\text{MEP} = \frac{w_{net}}{v_1 - v_2} = \frac{w_{net}}{v_1 - v_1/r} = \frac{w_{net}}{v_1(1 - 1/r)}$$

where

$$v_1 = \frac{RT_1}{P_1} = \frac{(0.287 \text{ kPa·m}^3\text{/kg·K})(290 \text{ K})}{100 \text{ kPa}} = 0.8323 \text{ m}^3\text{/kg}$$

Thus,

$$\text{MEP} = \frac{418.17 \text{ kJ/kg}}{(0.8323 \text{ m}^3\text{/kg})\left(1 - \frac{1}{8}\right)}\left(\frac{1 \text{ kPa·m}^3}{1 \text{ kJ}}\right) = \mathbf{574\ kPa}$$

(*e*) The total air mass taken by all four cylinders when they are charged is

$$m = \frac{V_d}{v_1} = \frac{0.0016 \text{ m}^3}{0.8323 \text{ m}^3\text{/kg}} = 0.001922 \text{ kg}$$

The net work produced by the cycle is

$$W_{net} = mw_{net} = (0.001922 \text{ kg})(418.17 \text{ kJ/kg}) = 0.8037 \text{ kJ}$$

That is, the net work produced per thermodynamic cycle is 0.8037 kJ/cycle. Noting that there are two revolutions per thermodynamic cycle ($n_{rev} = 2$ rev/cycle) in a four-stroke engine (or in the ideal Otto cycle including intake and exhaust strokes), the power produced by the engine is determined from

$$\dot{W}_{net} = \frac{W_{net}\dot{n}}{n_{rev}} = \frac{(0.8037 \text{ kJ/cycle})(4000 \text{ rev/min})}{2 \text{ rev/cycle}}\left(\frac{1 \text{ min}}{60 \text{ s}}\right) = \mathbf{26.8\ kW}$$

Discussion If we analyzed a two-stroke engine operating on an ideal Otto cycle with the same values, the power output would be calculated as

$$\dot{W}_{net} = \frac{W_{net}\dot{n}}{n_{rev}} = \frac{(0.8037 \text{ kJ/cycle})(4000 \text{ rev/min})}{1 \text{ rev/cycle}}\left(\frac{1 \text{ min}}{60 \text{ s}}\right) = 53.6 \text{ kW}$$

Note that there is one revolution in one thermodynamic cycle in two-stroke engines.

9.6 Diesel Cycle: The Ideal cycle for Compression-Ignition Engines

- In diesel engines, only air is compressed during the compression stroke, eliminating the possibility of autoignition (engine knock).
- Therefore, diesel engines can be designed to operate at much higher compression ratios than SI engines, typically between 12 and 24.

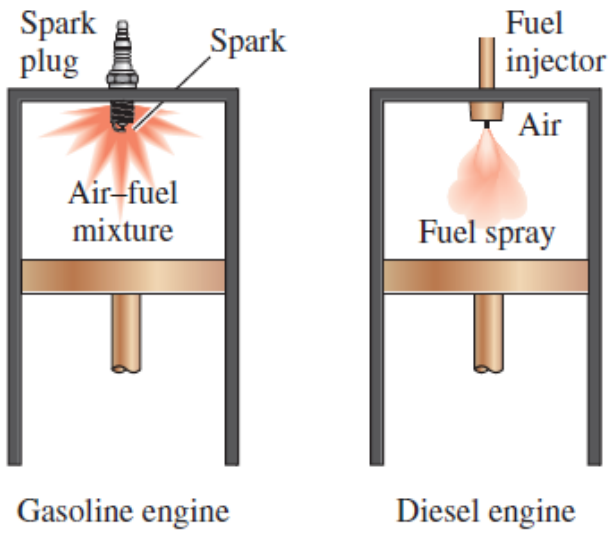


FIGURE 9–20
In diesel engines, the spark plug is replaced by a fuel injector, and only air is compressed during the compression process.

Process	Description
1-2	Isentropic Compression
2-3	Constant-Pressure Heat Addition
3-4	Isentropic Expansion
4-1	Constant-Volume Heat Rejection

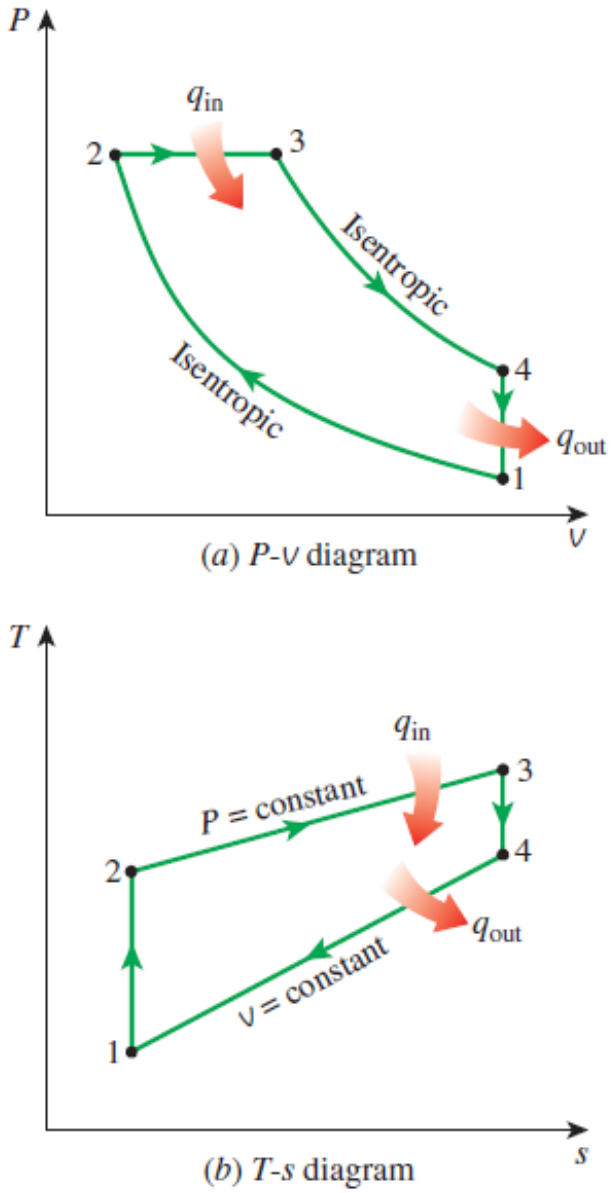
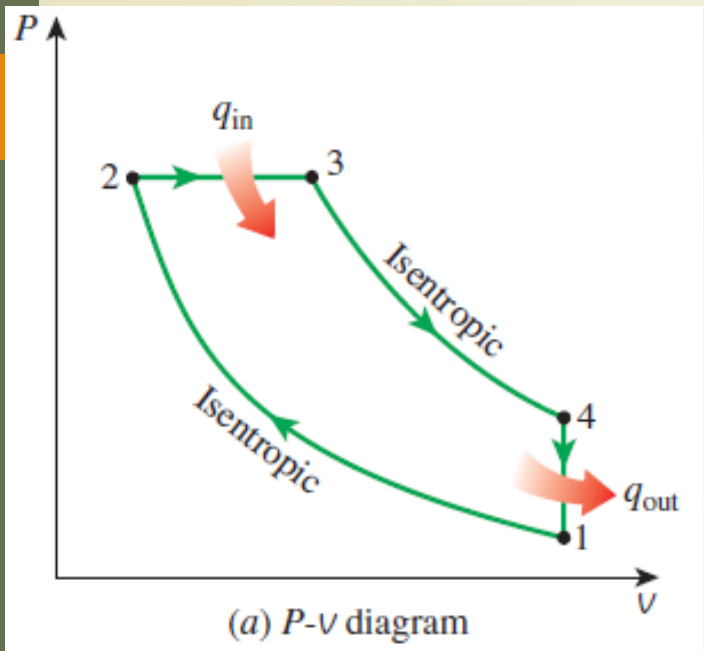
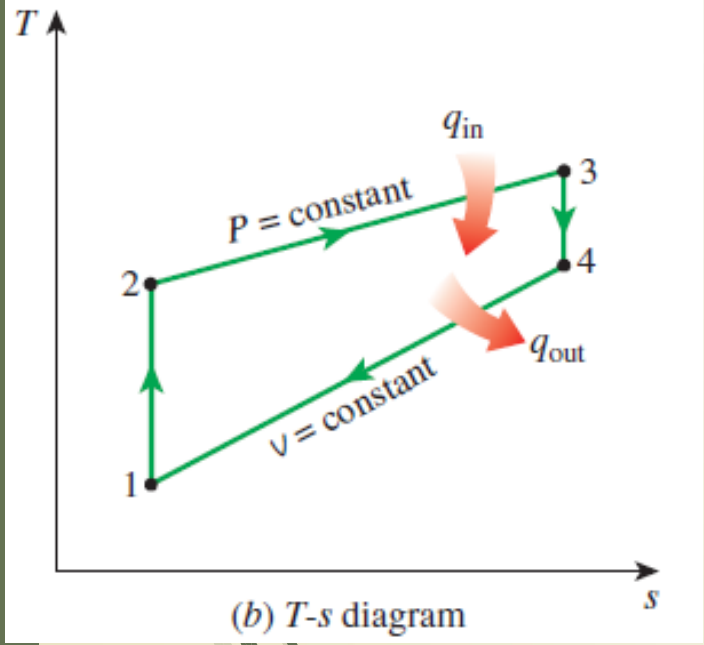


FIGURE 9–21
T-s and *P-v* diagrams for the ideal Diesel cycle.

9.6 Diesel Cycle: The Ideal cycle for Compression-Ignition Engines (2)



(a) P-v diagram



(b) T-s diagram

$$q_{in} - w_{b,out} = u_3 - u_2$$

$$q_{in} = P_2(v_3 - v_2) + (u_3 - u_2)$$

$$= h_3 - h_2 = c_p\,(T_3 - T_2) \qquad (9.9)$$

$$-q_{out} = u_1 - u_4$$

$$q_{out} = u_4 - u_1 = c_v(T_4 - T_1) \qquad (9.10)$$

- The thermal efficiency of the ideal Diesel cycle under the cold-air-standard assumptions becomes

$$\eta_{th,Diesel} = \frac{w_{net}}{q_{in}} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{T_4 - T_1}{k\,(T_3 - T_2)} = 1 - \frac{T_1\left(T_4/T_1 - 1\right)}{k\,T_2\left(T_3/T_2 - 1\right)}$$

9.6 Diesel Cycle: The Ideal cycle for Compression-Ignition Engines (3)

$$\frac{P_3V_3}{T_3} = \frac{P_2V_2}{T_2} \quad \text{where } P_3 = P_2$$

$$\frac{T_3}{T_2} = \frac{V_3}{V_2} = r_c$$

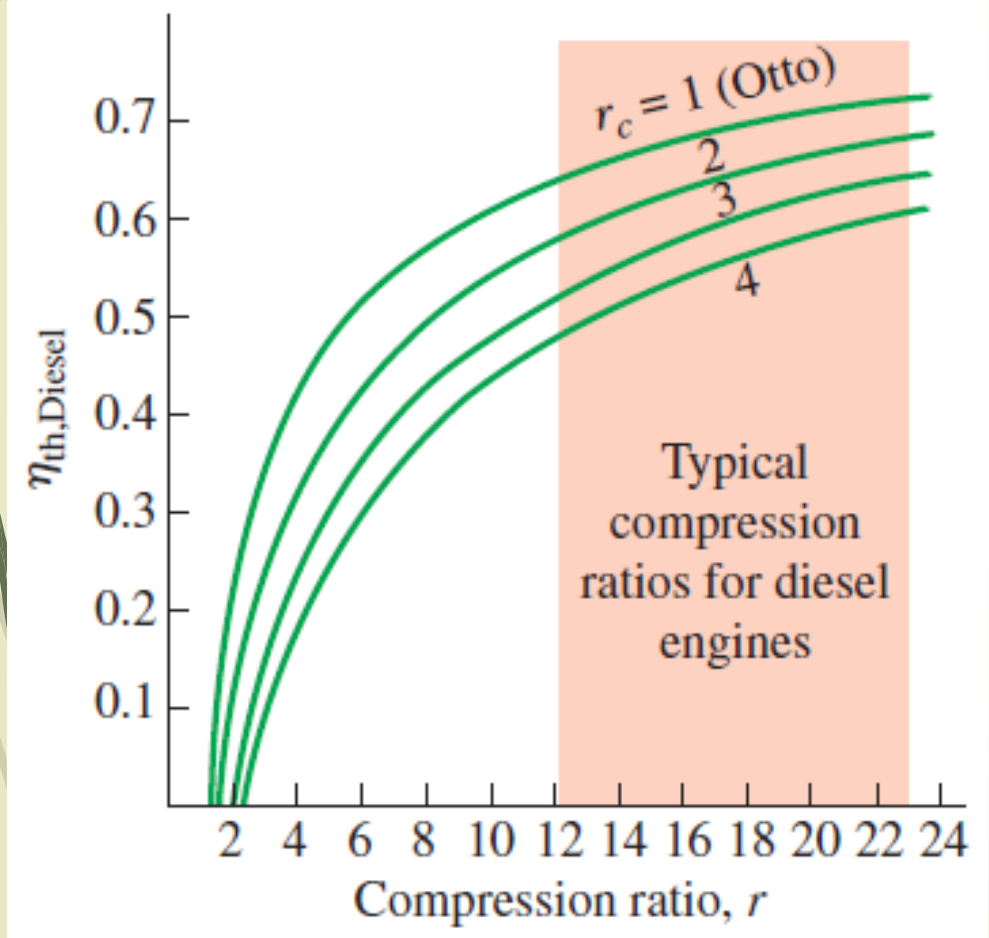
Where r_c is called the **cutoff ratio** defined as V_3 / V_2 and is a measure of the duration of the heat addition at constant pressure.

$$\frac{T_4}{T_1} = \frac{T_3}{T_2}\left(\frac{v_3/v_4}{v_2/v_1}\right)^{k-1} = \frac{v_3}{v_2}\left(\frac{v_3}{v_2}\right)^{k-1} = \left(\frac{v_3}{v_2}\right)^{k} = r_c^k$$

$$\eta_{th,\,Diesel} = 1 - \frac{1}{k}\frac{T_1(T_4/T_1 - 1)}{T_2(T_3/T_2 - 1)} = 1 - \frac{1}{k}\frac{T_1}{T_2}\frac{r_c^k - 1}{(r_c - 1)}$$

$$= 1 - \frac{1}{r^{k-1}}\left[\frac{r_c^k - 1}{k(r_c - 1)}\right]$$

9.6 Diesel Cycle: The Ideal cycle for Compression-Ignition Engines (4)



When $r_c > 1$ for a fixed r, $\eta_{th,\,Diesel} < \eta_{th,\,Otto}$

$$r_{Diesel} > r_{Otto}$$

$$\eta_{th,\,Diesel} > \eta_{th,\,Otto}$$

Thermal efficiency of the ideal Diesel cycle as a function of compression and cutoff ratios (k=1.4).

9.8 Brayton Cycle: The Ideal Cycle for Gas-Turbine Engines

- The Brayton cycle is the air-standard ideal cycle approximation for the gas-turbine engine.
- This cycle differs from the Otto and Diesel cycles in that the processes making the cycle occur in open systems or control volumes.

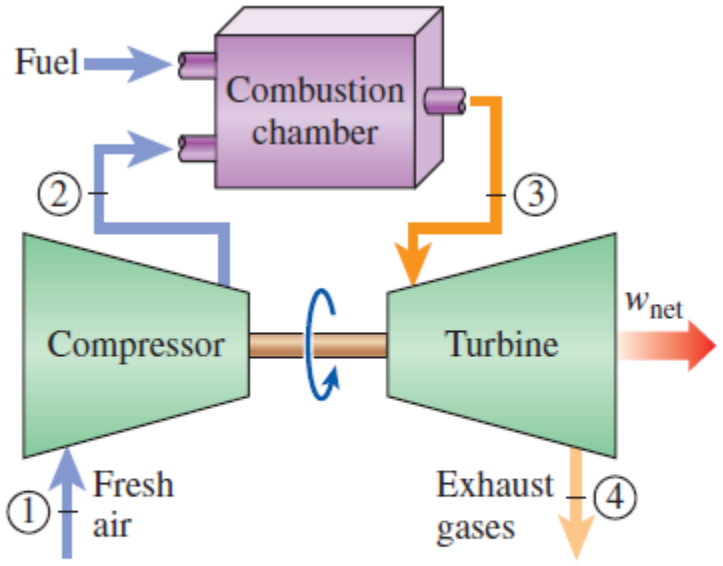


FIGURE 9–29
An open-cycle gas-turbine engine.

- Therefore, an open system, steady-flow analysis is used to determine the heat transfer and work for the cycle.
- We assume the working fluid is air and the specific heats are constant and will consider the cold air standard cycle.

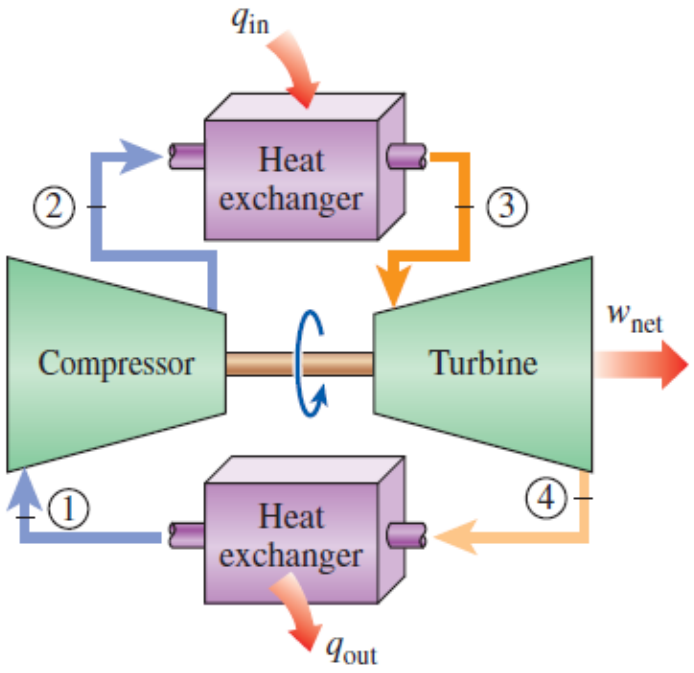


FIGURE 9–30
A closed-cycle gas-turbine engine.

Various engine types:
Turbofan, Propjet, Ramjet, Sacramjet, Rocket

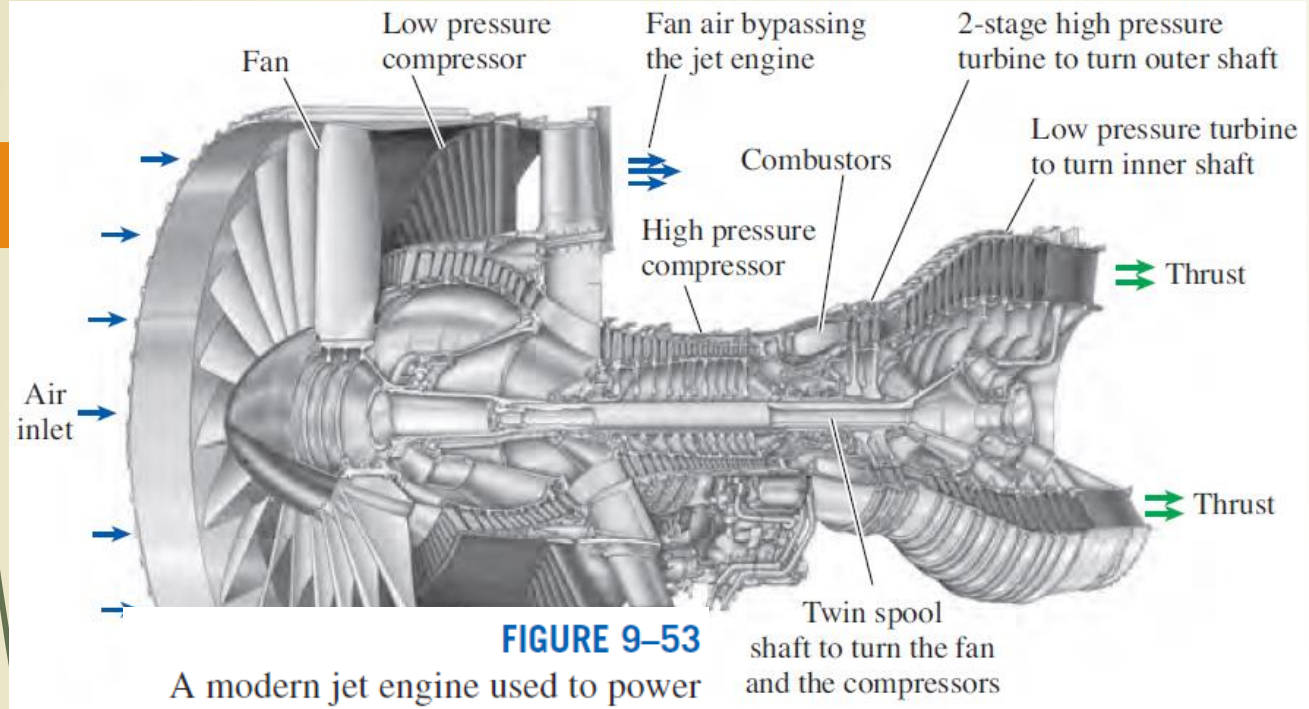


FIGURE 9–53
A modern jet engine used to power Boeing 777 aircraft. This is a Pratt & Whitney PW4084 turbofan capable of producing 84,000 pounds of thrust. It is 4.87 m (192 in) long, has a 2.84 m (112 in) diameter fan, and it weighs 6800 kg (15,000 lbm).

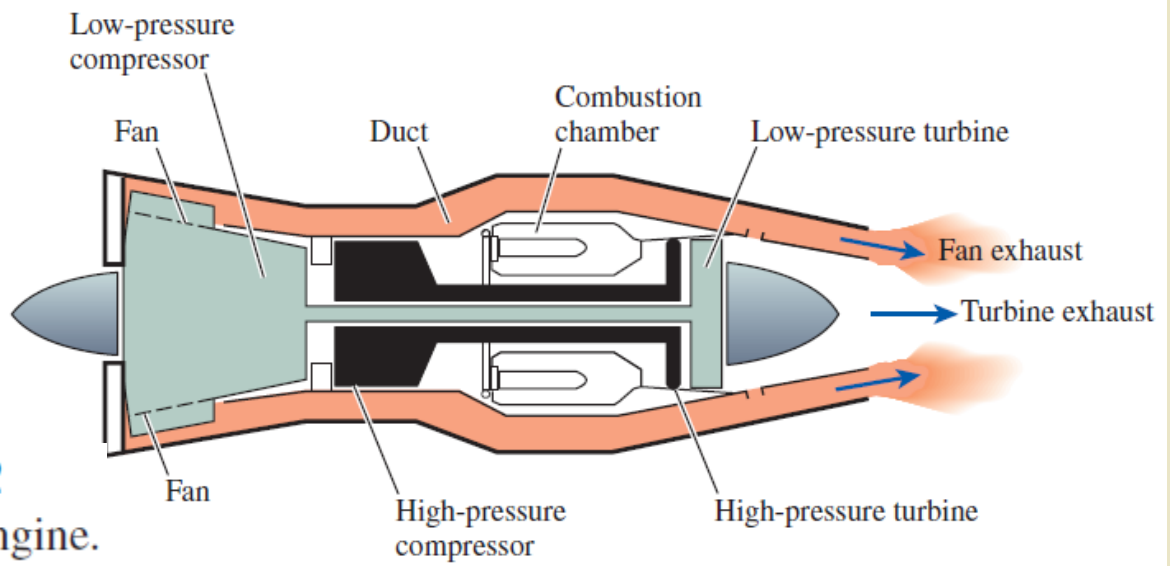


FIGURE 9–52
A turbofan engine.

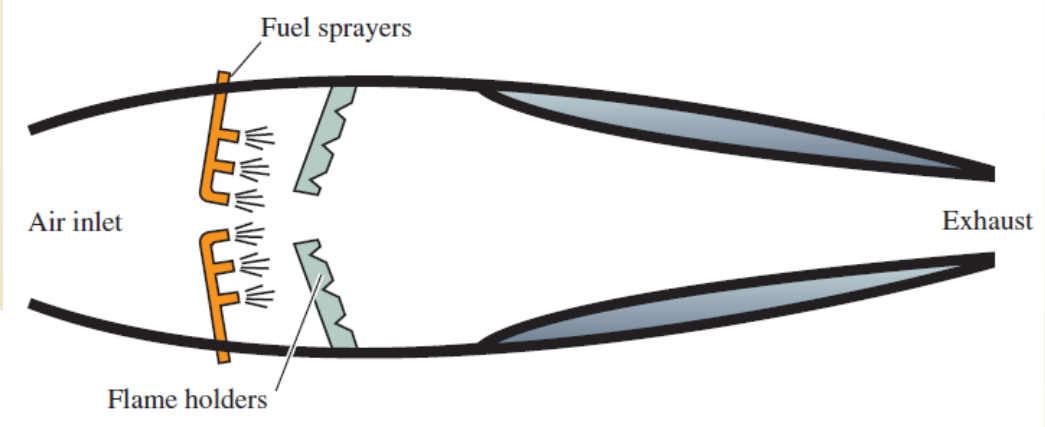


FIGURE 9–55
A ramjet engine.

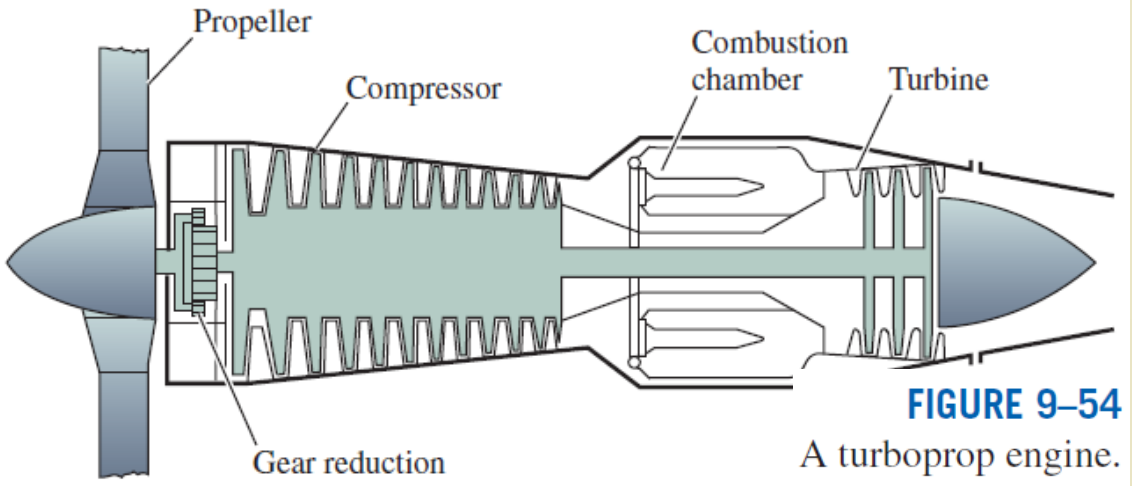


FIGURE 9–54
A turboprop engine.

9.8 Brayton Cycle: The Ideal Cycle for Gas-Turbine Engines (2)

- The combustion process is replaced by a constant-pressure heat-addition process from an external source, and the exhaust process is replaced by a constant-pressure heat-rejection process to the ambient air.

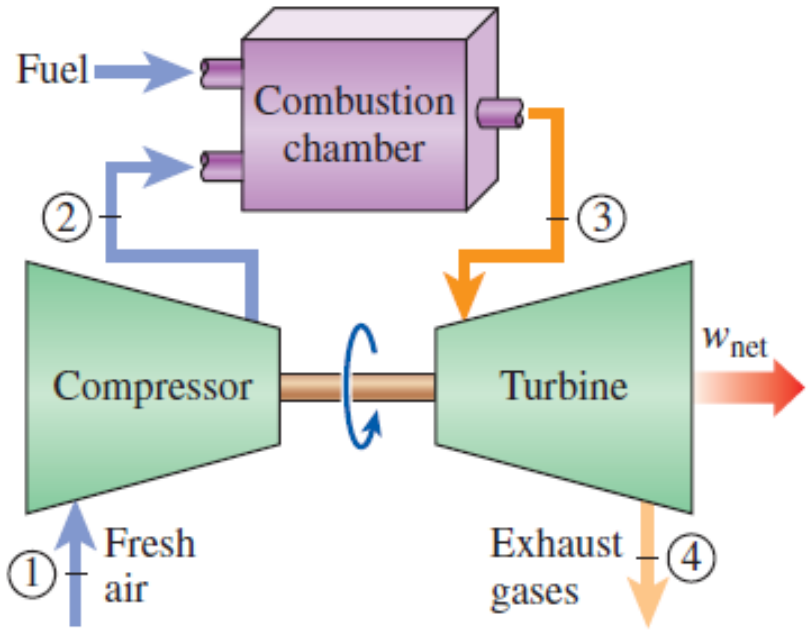


FIGURE 9–29
An open-cycle gas-turbine engine.

Process	Description
1-2	Isentropic Compression (in a compressor)
2-3	Constant-Pressure Heat Addition
3-4	Isentropic Expansion (in a turbine)
4-1	Constant-Pressure Heat Rejection

$$(q_{in} - q_{out}) + (w_{in} - w_{out}) = h_{exit} - h_{inlet}$$

$$q_{in} = h_3 - h_2 = c_p(T_3 - T_2)$$

$$q_{out} = h_4 - h_1 = c_p(T_4 - T_1)$$

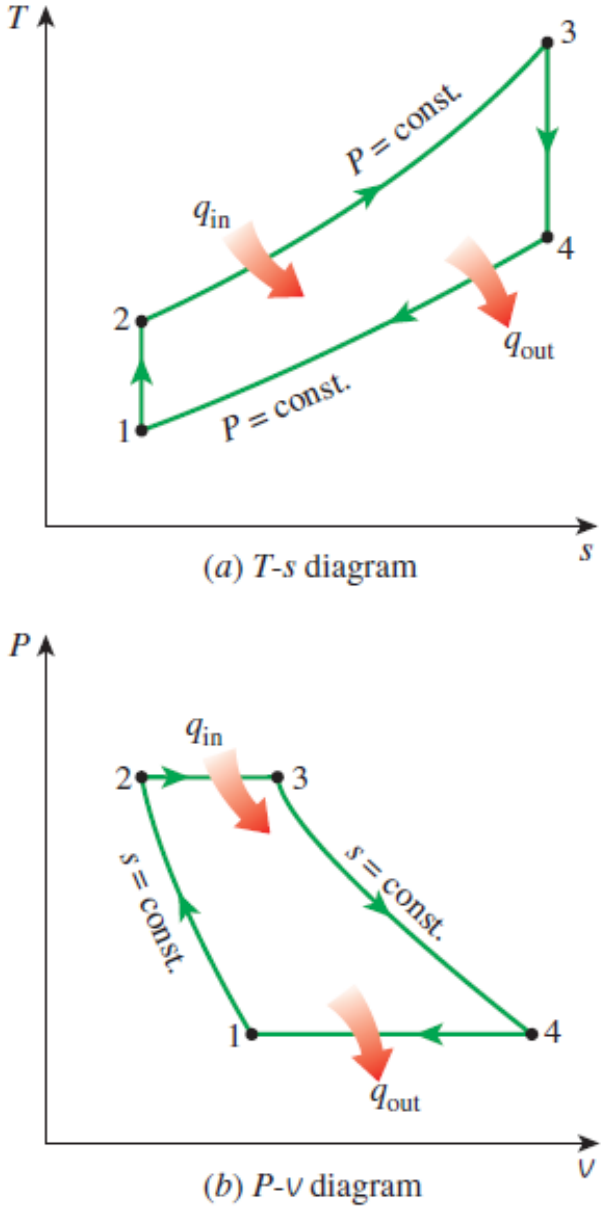


FIGURE 9–31
T-s and *P-v* diagrams for the ideal Brayton cycle.

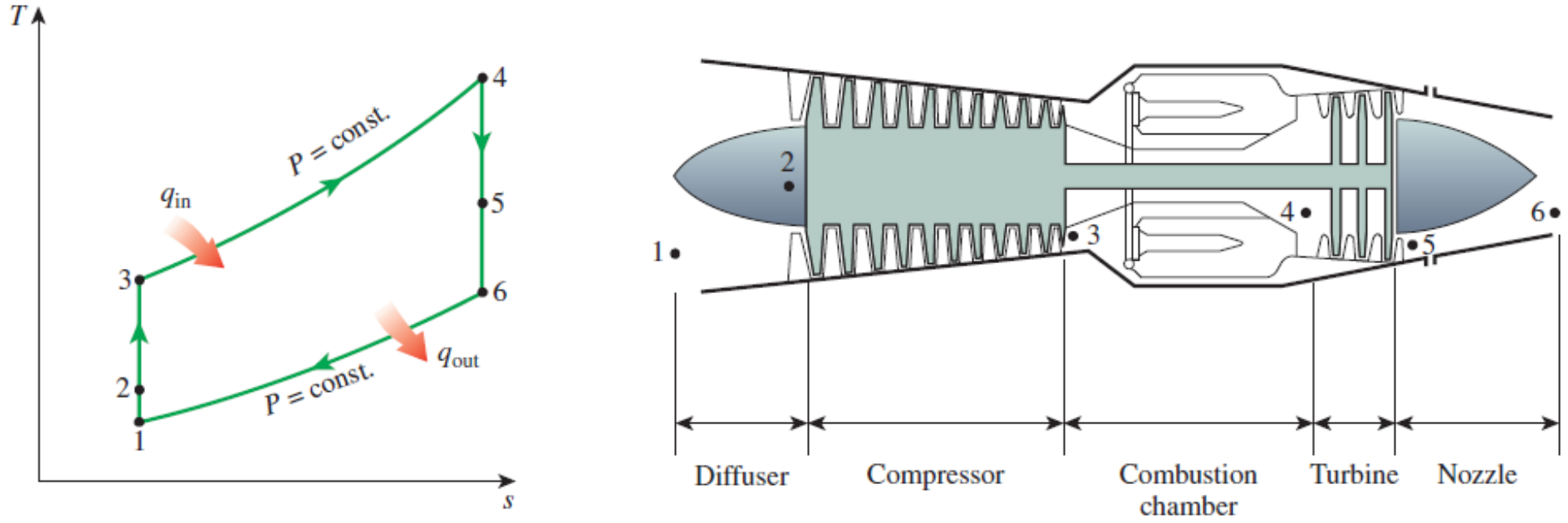


FIGURE 9–48
Basic components of a turbojet engine and the *T-s* diagram for the ideal turbojet cycle.

FIGURE 9–47
In jet engines, the high-temperature and high-pressure gases leaving the turbine are accelerated in a nozzle to provide thrust.

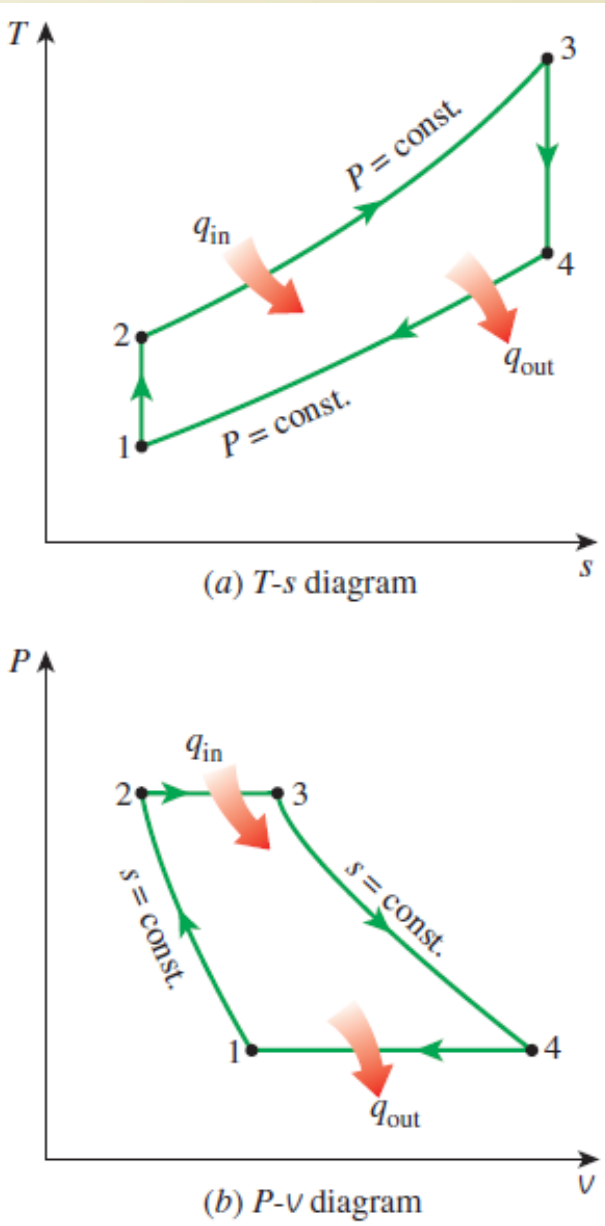


FIGURE 9–31
T-s and *P-v* diagrams for the ideal Brayton cycle.

9.8 Brayton Cycle: The Ideal Cycle for Gas-Turbine Engines (3)

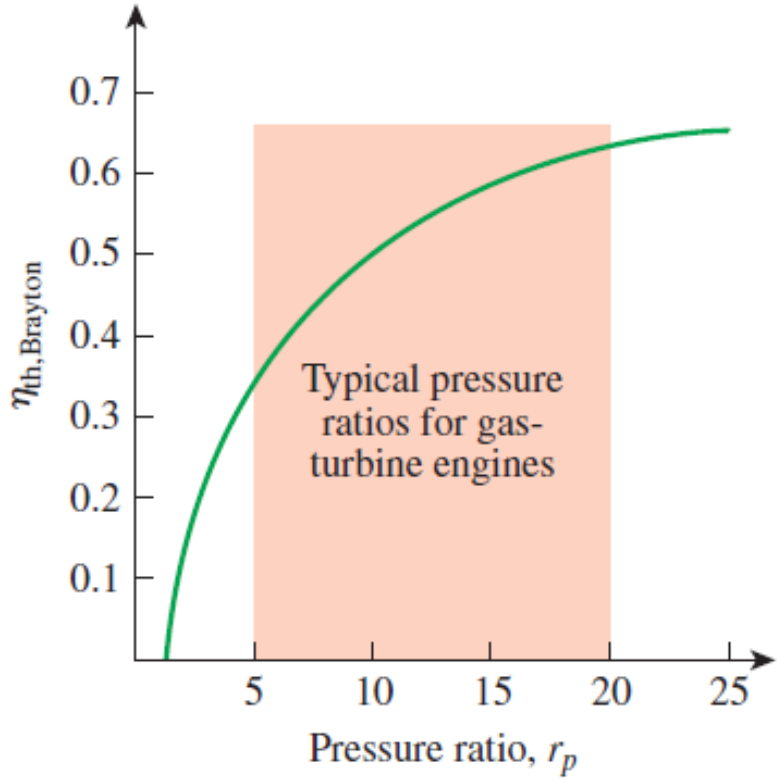


FIGURE 9–32
Thermal efficiency of the ideal Brayton cycle as a function of the pressure ratio.

$$\eta_{th,\,Brayton} = \frac{w_{net}}{q_{in}} = 1 - \frac{\dot{Q}_{out}}{\dot{Q}_{in}} = 1 - \frac{q_{out}}{q_{in}}$$

$$= 1 - \frac{C_p(T_4 - T_1)}{C_p(T_3 - T_2)} = 1 - \frac{(T_4 - T_1)}{(T_3 - T_2)} = 1 - \frac{T_1(T_4/T_1 - 1)}{T_2(T_3/T_2 - 1)}$$

- Processes 1-2 and 3-4 are isentropic, and $P_2 = P_3$ and $P_4 = P_1$. Thus,

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{(k-1)/k} = \left(\frac{P_3}{P_4}\right)^{(k-1)/k} = \frac{T_3}{T_4}$$

$$\therefore \frac{T_2}{T_1} = \frac{T_3}{T_4} \Rightarrow \frac{T_4}{T_1} = \frac{T_3}{T_2}$$

$$\eta_{th,\,Brayton} = 1 - \frac{T_1}{T_2} = 1 - \frac{1}{r_p^{(k-1)/k}}$$

Where $r_p = \dfrac{P_2}{P_1}$ is the **pressure ratio**.

9.8 Brayton Cycle: The Ideal Cycle for Gas-Turbine Engines (4)

- The two major application areas of gas-turbine engines are **aircraft propulsion** and **electric power generation**.

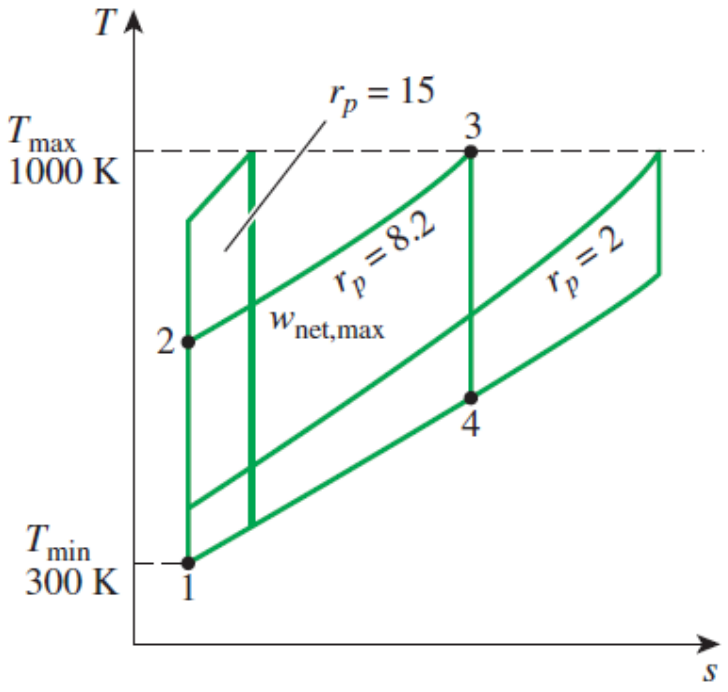


FIGURE 9–33
For fixed values of T_{min} and T_{max}, the net work of the Brayton cycle first increases with the pressure ratio, then reaches a maximum at $r_p = (T_{max}/T_{min})^{k/[2(k-1)]}$, and finally decreases.

- The highest temperature in the cycle is limited by the maximum temperature that the turbine blades can withstand.
- This also limits the pressure ratios that can be used in the cycle.
- The air in gas turbines supplies the necessary oxidant for the combustion of the fuel, and it serves as a coolant to keep the temperature of various components within safe limits.
- An air–fuel ratio of 50 or above is not uncommon

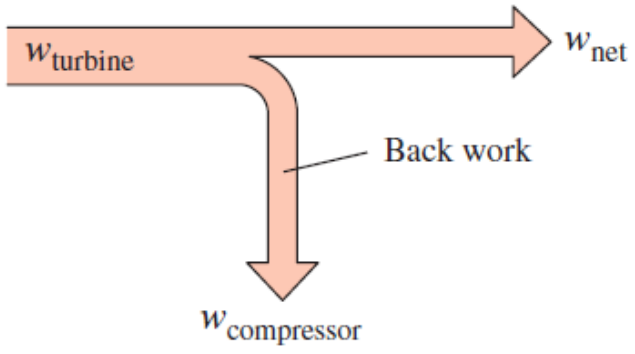


FIGURE 9–34
The fraction of the turbine work used to drive the compressor is called the back work ratio.

- The fraction of the turbine work used to drive the compressor is called the **back work ratio**.

Example 4

EXAMPLE 9–6 The Simple Ideal Brayton Cycle

A gas-turbine power plant operating on an ideal Brayton cycle has a pressure ratio of 8. The gas temperature is 300 K at the compressor inlet and 1300 K at the turbine inlet. Using the air-standard assumptions, determine (*a*) the gas temperature at the exits of the compressor and the turbine, (*b*) the back work ratio, and (*c*) the thermal efficiency.

SOLUTION A power plant operating on the ideal Brayton cycle is considered. The compressor and turbine exit temperatures, back work ratio, and thermal efficiency are to be determined.

Assumptions **1** Steady operating conditions exist. **2** The air-standard assumptions are applicable. **3** Kinetic and potential energy changes are negligible. **4** The variation of specific heats with temperature is to be considered.

Analysis The *T-s* diagram of the ideal Brayton cycle described is shown in Fig. 9–36. We note that the components involved in the Brayton cycle are steady-flow devices.

(*a*) The air temperatures at the compressor and turbine exits are determined from isentropic relations:

Process 1–2 (isentropic compression of an ideal gas):

$$T_1 = 300\text{ K} \rightarrow h_1 = 300.19\text{ kJ/kg}$$
$$P_{r1} = 1.386$$
$$P_{r2} = \frac{P_2}{P_1}P_{r1} = (8)(1.386) = 11.09 \rightarrow T_2 = \mathbf{540\ K} \qquad \text{(at compressor exit)}$$
$$h_2 = 544.35\text{ kJ/kg}$$

Process 3–4 (isentropic expansion of an ideal gas):

$$T_3 = 1300\text{ K} \rightarrow h_3 = 1395.97\text{ kJ/kg}$$
$$P_{r3} = 330.9$$
$$P_{r4} = \frac{P_4}{P_3}P_{r3} = \left(\frac{1}{8}\right)(330.9) = 41.36 \rightarrow T_4 = \mathbf{770\ K} \qquad \text{(at turbine exit)}$$
$$h_4 = 789.37\text{ kJ/kg}$$

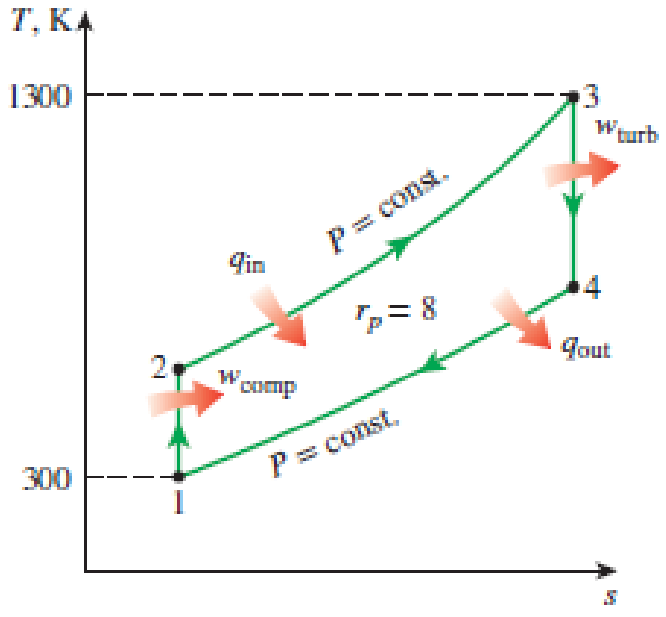


FIGURE 9–36
T-s diagram for the Brayton cycle discussed in Example 9–6.

Example 4

(*b*) To find the back work ratio, we need to find the work input to the compressor and the work output of the turbine:

$$w_{\text{comp,in}} = h_2 - h_1 = 544.35 - 300.19 = 244.16 \text{ kJ/kg}$$
$$w_{\text{turb,out}} = h_3 - h_4 = 1395.97 - 789.37 = 606.60 \text{ kJ/kg}$$

Thus,

$$r_{\text{bw}} = \frac{w_{\text{comp,in}}}{w_{\text{turb,out}}} = \frac{244.16 \text{ kJ/kg}}{606.60 \text{ kJ/kg}} = \mathbf{0.403}$$

That is, 40.3 percent of the turbine work output is used just to drive the compressor.

(*c*) The thermal efficiency of the cycle is the ratio of the net power output to the total heat input:

$$q_{\text{in}} = h_3 - h_2 = 1395.97 - 544.35 = 851.62 \text{ kJ/kg}$$
$$w_{\text{net}} = w_{\text{out}} - w_{\text{in}} = 606.60 - 244.16 = 362.4 \text{ kJ/kg}$$

Thus,

$$\eta_{\text{th}} = \frac{w_{\text{net}}}{q_{\text{in}}} = \frac{362.4 \text{ kJ/kg}}{851.62 \text{ kJ/kg}} = 0.426 \quad \text{or} \quad \mathbf{42.6\%}$$

The thermal efficiency could also be determined from

$$\eta_{\text{th}} = 1 - \frac{q_{\text{out}}}{q_{\text{in}}}$$

where

$$q_{\text{out}} = h_4 - h_1 = 789.37 - 300.19 = 489.2 \text{ kJ/kg}$$

Discussion Under the cold-air-standard assumptions (constant specific heat values at room temperature), the thermal efficiency would be, from Eq. 9–17,

$$\eta_{\text{th,Brayton}} = 1 - \frac{1}{r_p^{(k-1)/k}} = 1 - \frac{1}{8^{(1.4-1)/1.4}} = 0.448 \quad \text{or} \quad 44.8\%$$

which is sufficiently close to the value obtained by accounting for the variation of specific heats with temperature.

9.8 Brayton Cycle: The Ideal Cycle for Gas-Turbine Engines (5)

Deviation of Actual Gas-Turbine Cycles from Idealized Ones

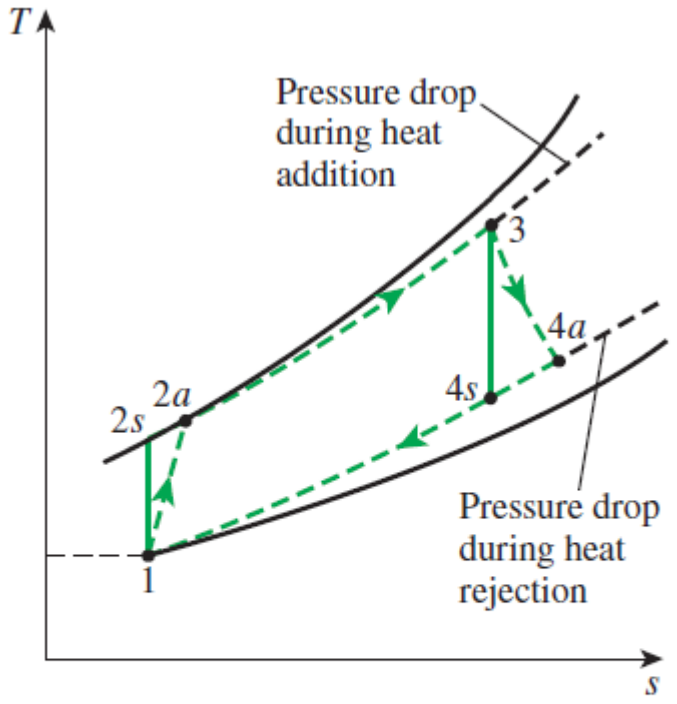


FIGURE 9–36
The deviation of an actual gas-turbine cycle from the ideal Brayton cycle as a result of irreversibilities.

- The actual gas-turbine cycle differs from the ideal Brayton cycle on several accounts.
  - some pressure drop during the heat-addition and heat-rejection processes is inevitable.
  - the actual work input to the compressor is more, and the actual work output from the turbine is less because of irreversibilities.
- The deviation of actual compressor and turbine behavior from the idealized isentropic behavior can be accurately accounted for by utilizing the isentropic efficiencies of the turbine and compressor as

$$\eta_C = \frac{w_s}{w_a} \cong \frac{h_{2s} - h_1}{h_{2a} - h_1} \qquad \eta_T = \frac{w_a}{w_s} \cong \frac{h_3 - h_{4a}}{h_3 - h_{4s}}$$

where states 2*a* and 4*a* are the actual exit states of the compressor and the turbine, respectively, and 2*s* and 4*s* are the corresponding states for the isentropic case

Example 5

EXAMPLE 9–7 An Actual Gas-Turbine Cycle

Assuming a compressor efficiency of 80 percent and a turbine efficiency of 85 percent, determine (*a*) the back work ratio, (*b*) the thermal efficiency, and (*c*) the turbine exit temperature of the gas-turbine cycle discussed in Example 9–6.

SOLUTION The Brayton cycle discussed in Example 9–6 is reconsidered. For specified turbine and compressor efficiencies, the back work ratio, the thermal efficiency, and the turbine exit temperature are to be determined.

Analysis (*a*) The *T-s* diagram of the cycle is shown in Fig. 9–38. The actual compressor work and turbine work are determined by using the definitions of compressor and turbine efficiencies, Eqs. 9–19 and 9–20:

Compressor:
$$w_{\text{comp,in}} = \frac{w_s}{\eta_C} = \frac{244.16 \text{ kJ/kg}}{0.80} = 305.20 \text{ kJ/kg}$$

Turbine:
$$w_{\text{turb,out}} = \eta_T w_s = (0.85)(606.60 \text{ kJ/kg}) = 515.61 \text{ kJ/kg}$$

Thus,

$$r_{\text{bw}} = \frac{w_{\text{comp,in}}}{w_{\text{turb,out}}} = \frac{305.20 \text{ kJ/kg}}{515.61 \text{ kJ/kg}} = \mathbf{0.592}$$

That is, the compressor is now consuming 59.2 percent of the work produced by the turbine (up from 40.3 percent). This increase is due to the irreversibilities that occur within the compressor and the turbine.

(*b*) In this case, air leaves the compressor at a higher temperature and enthalpy, which are determined to be

$$\begin{aligned} w_{\text{comp,in}} = h_{2a} - h_1 \rightarrow h_{2a} &= h_1 + w_{\text{comp,in}} \\ &= 300.19 + 305.20 \\ &= 605.39 \text{ kJ/kg} \qquad (\text{and } T_{2a} = 598 \text{ K}) \end{aligned}$$

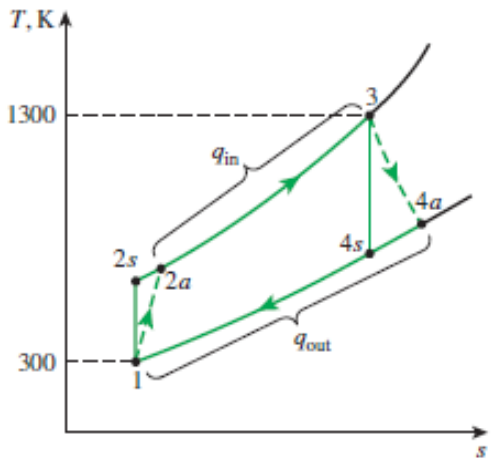


FIGURE 9–38
T-s diagram of the gas-turbine cycle discussed in Example 9–7.

Thus,

$$q_{\text{in}} = h_3 - h_{2a} = 1395.97 - 605.39 = 790.58 \text{ kJ/kg}$$
$$w_{\text{net}} = w_{\text{out}} - w_{\text{in}} = 515.61 - 305.20 = 210.41 \text{ kJ/kg}$$

and

$$\eta_{\text{th}} = \frac{w_{\text{net}}}{q_{\text{in}}} = \frac{210.41 \text{ kJ/kg}}{790.58 \text{ kJ/kg}} = 0.266 \quad \text{or} \quad \mathbf{26.6\%}$$

That is, the irreversibilities occurring within the turbine and compressor caused the thermal efficiency of the gas turbine cycle to drop from 42.6 to 26.6 percent. This example shows how sensitive the performance of a gas-turbine power plant is to the efficiencies of the compressor and the turbine. In fact, gas-turbine efficiencies did not reach competitive values until significant improvements were made in the design of gas turbines and compressors.

(*c*) The air temperature at the turbine exit is determined from an energy balance on the turbine:

$$\begin{aligned} w_{\text{turb,out}} = h_3 - h_{4a} \rightarrow h_{4a} &= h_3 - w_{\text{turb,out}} \\ &= 1395.97 - 515.61 \\ &= 880.36 \text{ kJ/kg} \end{aligned}$$

Then, from Table A–17,

$$T_{4a} = \mathbf{853 \text{ K}}$$

Discussion The temperature at turbine exit is considerably higher than that at the compressor exit ($T_{2a} = 598$ K), which suggests the use of regeneration to reduce fuel cost.

Thank You